

Edge-Native Data Mesh for Real-Time Tactical Deconfliction in 5G-Advanced U-space Traffic Management

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Abstract—The rapid growth of Uncrewed Aerial Systems (UAS) necessitates the evolution of UAS Traffic Management (UTM) frameworks that leverage 5G-advanced/6G connectivity and Multi-access Edge Computing (MEC) to satisfy stringent mission-critical safety requirements. While traditional aviation approaches are predominantly centralized, this paper proposes a decentralized MEC-enabled architecture combined with a sovereign data mesh governance model to improve real-time conflict responsiveness. Comprehensive simulations in a high-density 5G-enabled UTM scenario show a clear performance trade-off: far-edge processing achieves superior tactical end-to-end performance, whereas metro-edge deployments improve inter-domain migration efficiency through regional orchestration. Overall, the proposed federated MEC architectures enables the UTM operations by balancing local tactical safety with regional coordination, making it suitable for dense and time-critical aerial environments.

Keywords—UTM, 5G, MEC, Data Mesh, Very Low Level (VLL) Airspace

I. INTRODUCTION

The growing Uncrewed Aerial System (UAS) fleet is projected to reach 400,000 aircraft between 2035–2050, supporting wide range of commercial applications [1]. The projected increase from only a few thousand concurrently active UAS today, to the anticipated fleet size introduces unprecedented airspace density challenges. At Very Low Level (VLL) altitudes (approx. 150 m above ground level), conventional communication infrastructures struggle to support the real-time coordination, identification, and separation required for safe operations. The high density of UAS operations in VLL airspace demands more than connectivity alone. The UAS Traffic Management (UTM), central entity for trajectory planning, conflict detection, and monitoring is inherently both communication-intensive and computationally demanding, meaning that neither networking capacity nor processing power can be treated in isolation. Critically, these functions require an End-to-End (E2E) paradigm in which communication and computing are jointly provisioned: low-latency data exchange is only effective when paired with sufficient on-path computational resources to act upon that data in real time, making integrated Ultra-Reliable Low-Latency Communication (URLLC) and Computing, a fundamental requirement for viable UTM.

5G-advanced/6G through their native support for Multi-access Edge Computing (MEC) and cloud infrastructure, are uniquely positioned to deliver this integrated communication-and-computing substrate for safe, sustainable UTM in the VLL airspace [2]–[4]. While centralized cloud deployments support scalability for non-time-critical services, they cannot meet the stringent latency requirements of tactical (real-time) conflict detection and resolution operations. MEC addresses this by relocating computation to the network edge, providing the ultra-low latency and distributed responsiveness essential for safe, real-time UAS deconfliction in dense environments.

In the European context, this alignment is realized through *U-Space*, defined as a set of digital services and automated procedures governed by European aviation authority [5]. This set of digital services with critical data exchange between several key entities is designed to support safe, efficient, and secure access to airspace for large numbers of UAS. To operationalize these services, the aviation sector is adopting Service-Based Architectures (SBA) integrated with 5G/6G cores, enabling modular UTM with U-space functions such as network-assisted remote identification and real-time geo-awareness. However, a significant architectural mismatch remains: while current systems rely on centralized cloud designs for data exchange, the U-space vision demands a decentralized, sovereign ecosystem [6], [7]. This tension creates a critical performance gap, as centralized processing struggle to meet the reliability and data-sovereignty needs of distributed, edge-level decision-making.

Compounding this challenge, U-space involves a heterogeneous set of stakeholders, such as UAS operators, U-space digital Service Providers (SPs): U-Space Service Provider (USSP), Common Information Service Provider (CISP), Mobile Network Operator (MNO), and regulatory bodies; each owns fragments of operationally critical data on its own. The resulting fragmentation of data ownership, combined with high data intensity and ultra-low-latency demands, introduces significant governance and coordination challenges [8]. A *data mesh* [9] paradigm, wherein data ownership is decentralized to domain-specific producers who expose data as discoverable, interoperable products, directly addresses this fragmentation by enabling autonomous U-space entities to share and con-

sume data without centralized sovereignty bottlenecks. Applying data mesh principles within a MEC-enabled architecture therefore offers a principled response to both computational and governance dimensions of the problem.

To address the above-mentioned challenges, this paper proposes and evaluates a decentralized, data-mesh-driven MEC architectures, potentially ensuring scalable UAS coordination and seamless service migration across a federated U-space ecosystem. The core contributions are as following:

- *Network-Aware MEC for U-space:* We propose MEC-based architecture leveraging service-based principles, supporting both centralized and decentralized deployments for U-space, meeting stringent tactical deconfliction service requirements.
- *Federated Data Mesh Governance:* Application of data mesh principles to ensure secure, decentralized, and real-time interoperability among autonomous U-space entities.
- *Simulation-Based Validation:* An evaluation of MEC placement and service migration within a high traffic scenario, demonstrating Key Performance indicators (KPIs) for tactical responsiveness.

The remainder of this paper is organized as follows: Section II reviews the background and related work; Section III presents the proposed architecture, data mesh governance model, and system model for simulation ; Section IV analyses the KPIs; and Section V concludes with key findings and future work.

II. BACKGROUND

Technology enablers for mission-critical services: The next generation of 5G-Advanced and 6G mobile networks is designed to accommodate a wide range of vertical use cases through a cloud-native architecture that decouples network functions into modular, discoverable micro-services [10]. By extending the SBA to the network edge, MEC enables these mission-critical services to execute in close proximity to the end-user, thereby achieving the ultra-low latency required for high-mobility industrial applications [11]. Current research emphasizes the strategic hierarchical placement of functional entities: far-edge (Radio Access Network (RAN)-co-located), metro-edge (regional Point of Presence (PoP)), and central cloud to balance computational overhead with stringent latency targets. Within this distributed hierarchical framework, a key challenge is maintaining session continuity as UAS traverses high-speed trajectories, necessitating seamless service migration frameworks that minimize handover jitter and state-transfer delays [12]. Recent advancements in standardization in 3GPP Release 18/19, ETSI MEC Phase 4 addresses the required control and interoperability foundations for mobility support by introducing capabilities such as MEC Federator, and specialized Application Programming Interface (API) exposure capabilities - MEC 040, MEC 043 [13], [14]. Building upon these standardized capabilities, recent research focuses on optimizing their practical exploitation in dynamic mobility environments. In particular, policy-driven, heuristic based cloud-native approaches have been proposed to enhance service migration decisions by proactively adapting

to predicted UAS mobility patterns. These approaches aim to reduce service disruption and improve the timeliness of state synchronization across edge domains, thereby ensuring that safety-critical coordination functions operate on consistent and up-to-date system state information even during frequent inter-edge transitions [15].

Efforts towards 5G-integrated UTM: These technological enablers are being integrated into U-space, Europe's regulatory framework for the digitalization of VLL airspace, which defines progressive service levels - U1 provides basic foundational services such as UAS registration, identification, support for flight preparation. U2 provides initial operational services - flight trajectory planning, safety radius around each UAS trajectory and separation margin. U3 introduces advanced operational capabilities like real-time geo-awareness of the airspace, automated traffic management, tactical deconfliction (E2E process of detection, resolution, prevention). Finally, U4 enables highly automated and fully integrated U-space services for seamless integration and coordination with manned aviation for very high-density airspace.

Within this ecosystem, the USSP delivers safety-critical services to operators, while the CISP acts as the single source of truth for essential data exchange. While foundational initiatives have defined U-Space services U1 and U2, identified 5G slicing as essential for reliable control links, a critical sovereignty gap remains in multi-USSP environments, where data sharing for collision avoidance conflicts with proprietary interests [16]. Several ongoing validations highlight that current centralized cloud-based CISP platforms struggles with data sovereignty, especially during complex operations like deconfliction (U3) and often treat the underlying communication links and cloud-native functions as an ideal pipe, overlooking the dynamic latency and governance constraints of the network [17], [18].

Data governance in UTM: Beyond the technical presence of MEC infrastructure, the governance of data poses a more significant challenge than the subscribe-exchange messaging framework. Traditional centralized data management fails to address the data sovereignty, where safety-critical nature of UAS mission wanting high-speed access to data that is simultaneously subject to strict ownership and privacy constraints. Data Mesh paradigm treats data as a stand-alone product managed by the specific domain teams that generate it, effectively reducing backhaul congestion while maintaining the rigorous data privacy and localized control required for a federated U-space ecosystem [8], [19]. Building upon this decentralized layers, the following section proposes federated MEC architectures for tactical deconfliction service (U3), realizing the data mesh principles.

III. 5G-MEC ENABLED FEDERATED DATA ARCHITECTURE IN U-SPACE AND SIMULATION MODEL

Network-Aware MEC for U-space: The architecture in Figure 1, distinguishes between a centralized regional scenario realized at the metro-edge and a decentralized, localized model at the far-edge MEC for tactical deconfliction. The real-time geo-awareness is provided by the instance of CISP hosted at

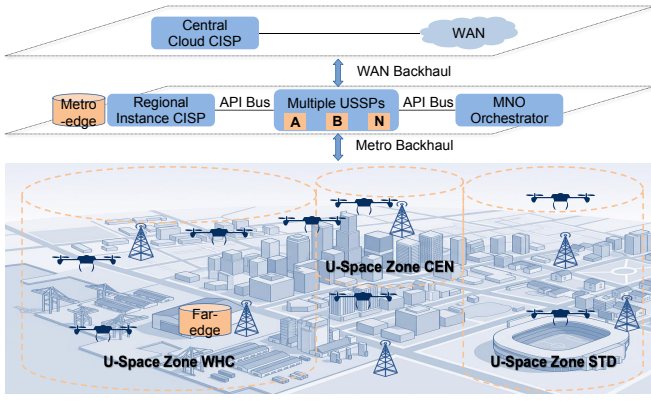


Figure 1. U-space Infrastructure with MEC (decentralized and centralized modes).

the MEC. In this simplified illustration, far-edge MEC node is co-located with base stations to provide ultra-low latency and immediate access to radio network data services for specific zones such as the warehouse (WHC), city centre (CEN), or stadium (STD). In contrast, the metro-edge serves as a regional *sweet spot*, hosting the regional instance of MEC, also MNO orchestrator to manage wide coverage across multiple 5G tiles with lower operational overhead than fragmented far-edge nodes. In this environment, the CISP owns the underlying infrastructure. The real-time geo-awareness is provided by the instance of CISP hosted at the MEC. While multiple USSPs (named as A, B, and so on up-to N) subscribes to the services to manage their respective UAS fleets via the metro, Wide Area Network (WAN) backhaul. This distribution allows for centralized strategic oversight while enabling the local autonomy required for high-density operations, though the shift toward decentralization raises critical data governance issues regarding ownership and verification across multi-actor domains. To ensure safety and service continuity, each U-space entity fulfils a distinct functional role. The MNO orchestrator manages micro-service lifecycles, routes traffic locally to MEC applications or outward to the central cloud. During tactical deconfliction, safety-critical data is exchanged via a high-capacity, API bus, a standardized inter-service communication layer for seamless safety-critical data exchange between central, metro and far-edge entities. This federated approach allows USSPs to process telemetry and geo-awareness as data products in real-time, bypassing the latency overhead of remote cloud round-trip processing. This provides the situational awareness required for immediate conflict resolution, the increased coordination overhead in high-density scenarios necessitates strict governance to ensure safety.

Federated Data Mesh Governance: To resolve the data sovereignty, governance and scalability challenges of decentralized MEC, we implement data mesh principles to federate data management as in Figure 2:

- **Domain-oriented ownership:** U-space stakeholders such as UAS operators, USSPs, and the CISP each own and manage their specific data products, such as mission

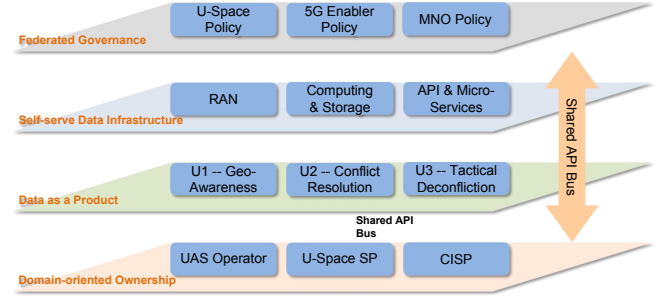


Figure 2. Data Mesh principle for U-Space.

telemetry, authorizations, rather than consolidating them into a monolithic central registry.

- **Data as a product:** Each entity publishes consumable, high-fidelity feeds such as geo-awareness or flight intent via standardized schemas and service-level agreements, treating data as a discoverable service.
- **Self-serve data infrastructure:** The standardized communication platform for publishing and subscribing to cross-domain products with minimal overhead.
- **Federated governance:** Global security policies and interoperability standards are enforced across all participants to ensure the integrity of real-time exchanges.

Sovereign data mesh ensures decentralized decision-making through domain-oriented ownership, allowing USSPs to control their tactical intent while subscribing to CISP-provided geo-awareness data. This replaces monolithic commands with federated, MEC-hosted logic that provides a verifiable audit trail for safety manoeuvres. Additionally, during transitions to non-U-space, the MNO orchestrator leverages self-serve infrastructure for seamless handovers, while the USSP maintains a safety buffer by publishing telemetry as a data product. Therefore, this approach preserves data sovereignty and mission history, enabling USSPs to manage localized billing and logs even beyond regulated zones.

System Simulation Model: To evaluate the performance of the proposed architecture, a Python-based simulation environment was developed to replicate a high-density, 5G/6G enabled U-space industrial-warehouse zone encompassing a 5 km operational area. The simulation models a peak load of 100 active UAS (multi-copters) flying generally up-to 120 m from the earth's surface in their recurrent pre-planned trajectory. The active UASs are randomly managed by five distinct USSPs that join the network at timed intervals of 2, 7, 10, and 15 minutes to simulate realistic traffic scaling. The central cloud CISP computing node is present to plan and approve the flight operations in this simulated area. We focus on real-time deconfliction (U3), defined as the E2E process in which conflicts between two or more UASs are detected, corresponding resolution actions are executed, and nominal operations are restored indicated by updated geo-awareness broadcast. Therefore, the study compares two primary architectural configurations: Scenario A (decentralized far-edge), and Scenario B (centralized metro-edge) respectively. The tac-

tical deconfliction must adhere to URLLC budget for enabling safe operations. It is important to note that this study focuses primarily on the comparative analysis of architectural designs and decision; therefore, complex ripple effects, stochastic environmental uncertainties, and other operational variability were not integrated into the current simulation.

In Scenario A, where USSPs are subscribed to the far-edge node present at the industrial zone as in Figure 1, it get real-time geo-awareness of other active USSPs operating in this region. When a conflict between two UASs is detected, the far-edge MEC co-located with the 5G base station immediately signals conflict and resolution action to the respective USSPs. The conflict resolution logic is processed entirely at the far-edge, with the optimized solution delivered directly to the involved USSPs to initiate corrective manoeuvres. These corrective manoeuvres are adjustments to altitude, speed, or heading, and executed in strict accordance with aviation standards tailored to UAS performance characteristics. In this model, the metro-edge is bypassed for immediate tactical decision loops, as the localized infrastructure provides sufficient autonomy to maintain safety without the latency overhead of higher-tier coordination.

In Scenario B (Figure 1), the architecture transitions to a higher layer where tactical deconfliction significantly extending the communication path. In this case, telemetry data from the involved UAS must traverse the metro backhaul to reach the corresponding USSP instances. This dependence on geographically distant processing introduces additional latency, which becomes particularly critical in high-density environments. Moreover, the metro-edge layer is regionally centralized and thus does not offer the fine-grained, tactical resolution required during deconfliction. Instead, it informs the respective USSPs and allows them to negotiate conflict resolution among themselves. Instead of directly computing and enforcing conflict resolution, it disseminates situational awareness to the respective USSPs, allowing them to negotiate resolution strategies. In this context, distributed negotiation is more suitable, as it leverages the local operational knowledge of each USSP, avoids overloading the centralized metro-edge with complex optimization tasks, and mitigates bottlenecks inherent to centralized control, adhering to data mesh principles. The WAN backhaul transports reporting, traffic logging data, enabling the exchange of operational actions between USSPs and MEC instances and their subsequent delivery to the central CISP cloud. Beyond tactical execution, the metro-edge serves as the critical coordination layer for ensuring session continuity during inter-zonal transitions, such as a UAS moving from U-Space Zone WHC to Zone CEN. As the MNO orchestrator monitors the flight's trajectory toward a zone boundary, it proactively prepares the target far-edge MEC by pre-staging the subscriber's USSP and relevant data products. This regional oversight allows the metro-edge to synchronize the UAS and U-space zone state information across the API bus, ensuring that the UAS's safety radius and active tactical constraints are seamlessly handed over. Consequently, the metro-edge acts as a state-replication an-

chor that maintains a persistent sovereign data environment, allowing the UAS to traverse multiple localized 5G tiles while maintaining uninterrupted deconfliction services. Additionally, we envision WAN backhaul to the CISP central cloud also functions as a fallback in case of regional failures to avoid complete loss in communication within U-space. In Scenario B, where coordination is shifted to a regional data centre, necessitating WAN and metro backhaul traversals for every data exchange.

Key Performance Metrics: The performance is quantified through E2E latency T_{E2E} , Age of Information (AoI) $\Delta(t)$, and handover time T_{HO} of UAS operation between U-spaces.

Latency: It accounts for respective uplink/downlink propagation (which is assumed to be symmetrical in this work), internal processing time (T_{proc}), and synchronization delays. The 3GPP based general latency profile is represented as,

$$T_{E2E} = T_{Detect} + T_{Comm} + T_{MEC_Proc} + T_{Execute}, \quad (1)$$

where T_{Detect} and $T_{Execute}$ are the processing delays of tactical deconfliction workflow, T_{Comm} is the overall communication-related delay in 5G, T_{MEC_Proc} is the MEC computing delay. Here, the URLLC budget, which 3GPP defines as necessitating $T_{E2E} = 10$ ms for tactical safety and immediate conflict resolution. The propagation delay within 40 ms, far-edge delay up-to 10 ms margin and metro-edge margin up to 40 ms. Scenario A, E2E far-edge (FE) latency (L) (L_{FE}) is represented as below,

$$L_{FE} = 2 \times (T_{Comm} + T_{MEC_proc} + T_{USSP_proc}), \quad (2)$$

where 2 times denotes latency of involved USSPs, until the final successful deconfliction, T_{Comm} and T_{MEC_proc} as in Equation (1). The USSP processing delay T_{USSP_proc} includes resolution action to the corresponding UAS operator, successful completion, update to far-edge MEC and broadcast of new geo-awareness to the existing USSPs. The complete system is decentralized and federated to the operational geo-zone. The regional or central core layers are not invoked in this process. Meanwhile, in Scenario B, the E2E metro-edge (ME) latency L_{ME} is given in Equation (3), where significant backhaul delay (T_{BH}) and multi-step negotiation (T_{Neg}) penalize, as conflict resolution relies on the involved USSPs decision.

$$L_{ME} = 2 \times (T_{Comm} + T_{BH} + T_{MEC_proc} + T_{USSP_proc}) + T_{Neg}. \quad (3)$$

AoI of the tactical information: The exchange during tactical conflict detection, resolution is crucial. The AoI ($\Delta(t)$) for tactical deconfliction in the proposed architectures are evaluated based on $\Delta(t) = t - \max_{i:t_{u,i} \leq t}(t_{g,i})$, where, t is current observation of successful deconfliction time at the receiver (far-edge or metro-edge), $t_{u,i}$ is time at which the i -th update was received to the system, $t_{g,i}$ is the time at which that i -th update was generated by the UAS operators after the execution. The max considers most recent successfully received packet before time t .

Handover time between U-spaces: The handover time (T_{HO}) is defined as the total duration required to transfer the

UAS state and control information between U-spaces within same regional zones. In case of centralized Scenario B, the handover occurs in the intra-MEC manner, hence, the handover time is calculated referring to Equation (4a). While in Scenario A, far-edge setup requires a backhaul transfer between two different far-edge MECs, which adds complexity and delay during state transfers. This is because, as shown in Equation (4b), the metro-edge already holds the registry and the USSP instances for the entire region ($T_{state_transfer}$), the handover is often a logical re-assignment rather than a physical data migration (T_{sync}), with $T_{E2E} - T_{Comm}$ and T_{MEC_proc} delays.

$$T_{HO_ME} = T_{Comm} + T_{state_transfer} + T_{MEC_proc} + T_{sync}, \quad (4a)$$

$$T_{HO_FE} = T_{Comm} + T_{FE1_to_MEC} + T_{state_transfer} + T_{MEC_to_FE2} + T_{sync}. \quad (4b)$$

IV. RESULTS AND ANALYSIS

To evaluate the proposed architecture, we developed a comprehensive simulation of a high-density, 5G-enabled U-space environment. This section analyses the KPIs, including E2E latency, AoI, and handover time for the Scenario A and B. The Figure 3 illustrates the E2E latency fluctuations (upper graph) over a 100 minute simulation period for Scenario A (green) and B (orange) respectively. The associated statistics of Cumulative Distribution Function (CDF) in Figure 3 (bottom graph), further evaluates compliance with URLLC constraints over 10,000 simulation runs. Both scenarios maintain low latency during geo-awareness broadcasts; however, performance diverges (higher latency) during tactical conflict events (at 22 and 65 minutes). Scenario A, where deconfliction is resolved at the far-edge, has peak latency around 18 ms. In contrast, Scenario B exhibits higher peaks of up-to 67 ms due to USSP negotiation overhead. In Figure 3 (bottom graph), Scenario A achieves approximately 90% reliability within the URLLC budget with tail latency remaining below 20 ms even under conflict conditions. Scenario B, however, does not meet the strict budget, with baseline latency starting above 18 ms and extending toward 70 ms under peak load, making it more suitable for less time-critical operations.

Figure 4 presents AoI dynamics under linear ageing of tactical state information. Scenario A maintains consistently low information ageing (12–32 ms) due to proximity to the end users, ensuring highly up-to-date UAS position awareness. In contrast, Scenario B suffers from higher AoI values (52–72 ms), primarily due to backhaul and coordination delays. At typical UAS speeds (15 m/s), this translates to positional drift exceeding 1 m, introducing uncertainty in maintaining safety radius separation constraints. Therefore, Scenario A is a better architecture in providing a tactical safety capacity. However, Figure 5 demonstrates a reversal in performance during U-space migration, where the Scenario B provides a clear architectural advantage. Here, it demonstrates superior efficiency due to its regional orchestration capability, enabling faster handovers within a metro-edge cluster and

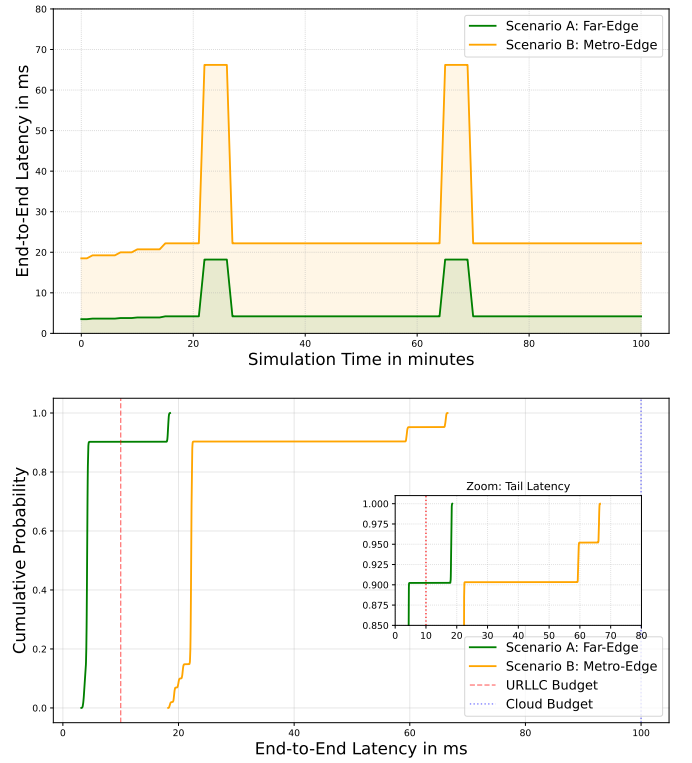


Figure 3. End-to-End latency for Scenario A (green) and Scenario B (orange) during real-time tactical deconfliction (single simulation) on the upper part, and CDF for Scenario A and B - bottom part.

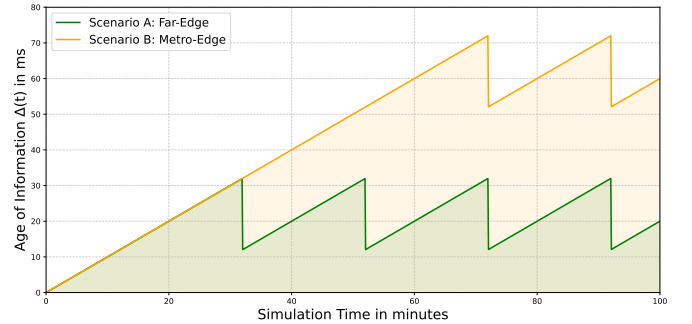


Figure 4. Age of Information during single simulation instance - Scenario A and B (single simulation).

remaining within the 100 ms migration cloud budget. Conversely, Scenario A experiences higher migration latency due to its reliance on WAN backhaul to central cloud services, leading to frequent budget violations in higher percentiles.

The quantitative comparison in Table I consolidates these findings by summarizing latency, AoI, migration efficiency, reliability, and governance across cloud-based state-of-the-art recommendation and the proposed architectures. It highlights the trade-off between both scenarios, where Scenario A provides superior real-time tactical performance with minimal latency and higher freshness. On the other hand, Scenario B offers improved migration efficiency and regional coordination advantages. Overall, the results highlight that no single

TABLE I. QUANTITATIVE KPI COMPARISON FOR STATE-OF-ART AND PROPOSED ARCHITECTURES FOR DECONFLICTION (U3)

KPI	Cloud (Recommended State-of-Art)	State-of-Art	Proposed Scenario A: Far-Edge	Proposed Scenario B: Metro-Edge
Tactical Latency	High (>100 ms); fails URLLC safety requirements.	URLLC	Best: Stable low (<15 ms).	Moderate (40–70 ms) due to backhaul (T_{BH}) and negotiation.
AoI	Poor (>150 ms); high risk of acting on stale UAS telemetry.		Best: High freshness; peaks at ~32 ms with 12 ms floor.	Moderate; floor limited to 52 ms by transport latency.
Migration Efficiency	Poor; complex global registry lookups via WAN.		Limited; high latency (~101 ms) due to backhaul.	Best: Seamless (<90 ms); due to regional view.
Reliability (95 Percentile)	Very Low; high jitter during network congestion.		High: below 20 ms even during conflict spikes.	Moderate: exceeds 65 ms during tactical deconflation.
Governance	Simplest: Single point of control/registry.		Complex: Fragmented data ownership across multiple FE nodes.	Optimal: Neutral regional orchestration via CISP mesh.

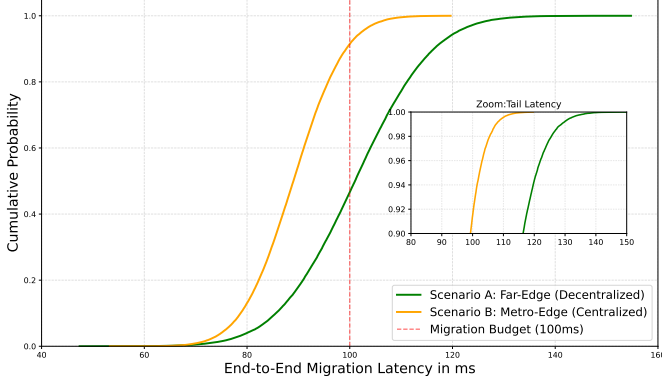


Figure 5. Migration Latency using Scenario A and B for a single UAS from one to another U-space within same regional location.

architecture dominates across all KPIs, but instead exhibits complementary strengths depending on operational context (tactical deconflation versus mobility management).

V. CONCLUSION

This paper proposes MEC-enabled architecture driven by data mesh principles for UTM to address the critical trade-off between latency, scalability, and federated governance in U-space as summarized in Table I. Our findings demonstrate that far-edge MEC provides the best performance for tactical deconflation due to its ultra-low latency, making it suitable for high-density, mission-critical environments such as industrial and logistics domains. In contrast, the metro-edge offers a balanced regional solution that reduces central cloud dependency while maintaining adequate coordination performance. The central CISP cloud remains essential for strategic, non-latency-critical functions such as planning and geo-awareness. By applying data mesh principles, the architecture enables distributed governance of data products across the edge–cloud continuum, balancing operational cost with interoperability requirements. Future work will extend the complex simulation framework to evaluate traffic prioritization and event-driven coordination under varying load conditions.

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